

$$+ P(\text{pass test}|\text{non-survivor})P(\text{non-survivor})$$

We know that  $P(\text{survivor}) = 1 - P(\text{non-survivor}) = 1 - 0.40 = 0.60$ . Therefore,  $P(\text{pass test}) = 0.55 = 0.85(0.60) + P(\text{pass test} | \text{non-survivor})(0.40)$ .

Thus,  $P(\text{pass test} | \text{non-survivor}) = [0.55 - 0.85(0.60)]/0.40 = 0.10$ .

B. We can calculate the probability that a company is a survivor as follows:

$$\begin{aligned} P(\text{survivor}|\text{pass test}) &= [P(\text{pass test}|\text{survivor})/P(\text{pass test})]P(\text{survivor}) \\ &= (0.85/0.55)0.60 = 0.927273 \end{aligned}$$

The information that a company passes the test causes you to update your probability that it is a survivor from 0.60 to approximately 0.927.

C. According to Bayes' formula,  $P(\text{non-survivor} | \text{fail test}) = [P(\text{fail test} | \text{non-survivor}) / P(\text{fail test})]P(\text{non-survivor}) = [P(\text{fail test} | \text{non-survivor})/0.45]0.40$ .

We can set up the following equation to obtain  $P(\text{fail test} | \text{non-survivor})$ :

$$\begin{aligned} P(\text{fail test}) &= P(\text{fail test}|\text{non-survivor})P(\text{non-survivor}) \\ &+ P(\text{fail test}|\text{survivor})P(\text{survivor}) \\ 0.45 &= P(\text{fail test}|\text{non-survivor})0.40 + 0.15(0.60) \end{aligned}$$

where  $P(\text{fail test} | \text{survivor}) = 1 - P(\text{pass test} | \text{survivor}) = 1 - 0.85 = 0.15$ . So,  $P(\text{fail test} | \text{non-survivor}) = [0.45 - 0.15(0.60)]/0.40 = 0.90$ .

Using this result with the previous formula, we find  $P(\text{non-survivor} | \text{fail test}) = [0.90/0.45]0.40 = 0.80$ . Seeing that a company fails the test causes us to update the probability that it is a non-survivor from 0.40 to 0.80.

D. A company passing the test greatly increases our confidence that it is a survivor. A company failing the test doubles the probability that it is a non-survivor. Therefore, the test appears to be useful.

3. An analyst estimates that 20 percent of high-risk bonds will fail (go bankrupt). If she applies a bankruptcy prediction model, she finds that 70 percent of the bonds will receive a "good" rating, implying that they are less likely to fail. Of the bonds that failed, only 50 percent had a "good" rating. Using Bayes' formula, what is the predicted probability of failure given a "good" rating? (Hint, let  $P(A)$  be the probability of failure,  $P(B)$  be the probability of a "good" rating,  $P(B | A)$  be the likelihood of a "good" rating given failure, and  $P(A | B)$  be the likelihood of failure given a "good" rating.)

- A. 5.7 percent
- B. 14.3 percent
- C. 28.6 percent

**Solution:**

B is correct. With Bayes' formula, the probability of failure given a "good" rating is

$$P(A|B) = \frac{P(B|A)}{P(B)}P(A)$$

where

$P(A) = 0.20$  = probability of failure

$P(B) = 0.70$  = probability of a "good" rating

$P(B | A) = 0.50$  = probability of a "good" rating given failure

With these estimates, the probability of failure given a "good" rating is